

Group & Ring Theory MATP33: Course Summary of Important Results

Simon Gustafsson

Lecture 1

Theorem 1 (1.10). *Let G be a group and X a set. Then:*

- (i) If X is a G -set, then the action of G on X induces a homomorphism $\varphi : G \rightarrow S_X$;*
- (ii) Any homomorphism $\varphi : G \rightarrow S_X$ induces an action of G on X .*

Proof. (i) Define $\varphi : G \rightarrow S_X$ by $(\varphi(a))(x) = a \cdot x$ for $a \in G$ and $x \in X$. We must first check that $\varphi(a) \in S_X$ for each $a \in G$, i.e. that the function $x \mapsto a \cdot x$ is a bijection $X \rightarrow X$.

- *Injective:* Suppose $a \cdot x = a \cdot y$. Then $x = a^{-1} \cdot (a \cdot x) = a^{-1} \cdot (a \cdot y) = y$.
- *Surjective:* For any $x \in X$, we have $a \cdot (a^{-1} \cdot x) = x$, so $x \in \text{im}(\varphi(a))$.

Hence $\varphi(a) \in S_X$. To see φ is a homomorphism, let $a, b \in G$. Then for all $x \in X$,

$$(\varphi(ab))(x) = (ab) \cdot x = a \cdot (b \cdot x) = (\varphi(a))((\varphi(b))(x)) = (\varphi(a)\varphi(b))(x).$$

(ii) Define $a \cdot x := (\varphi(a))(x)$. Then $(ab) \cdot x = (\varphi(ab))(x) = (\varphi(a)\varphi(b))(x) = a \cdot (b \cdot x)$, and $e \cdot x = (\varphi(e))(x) = x$. Hence X is a G -set. $\square$